



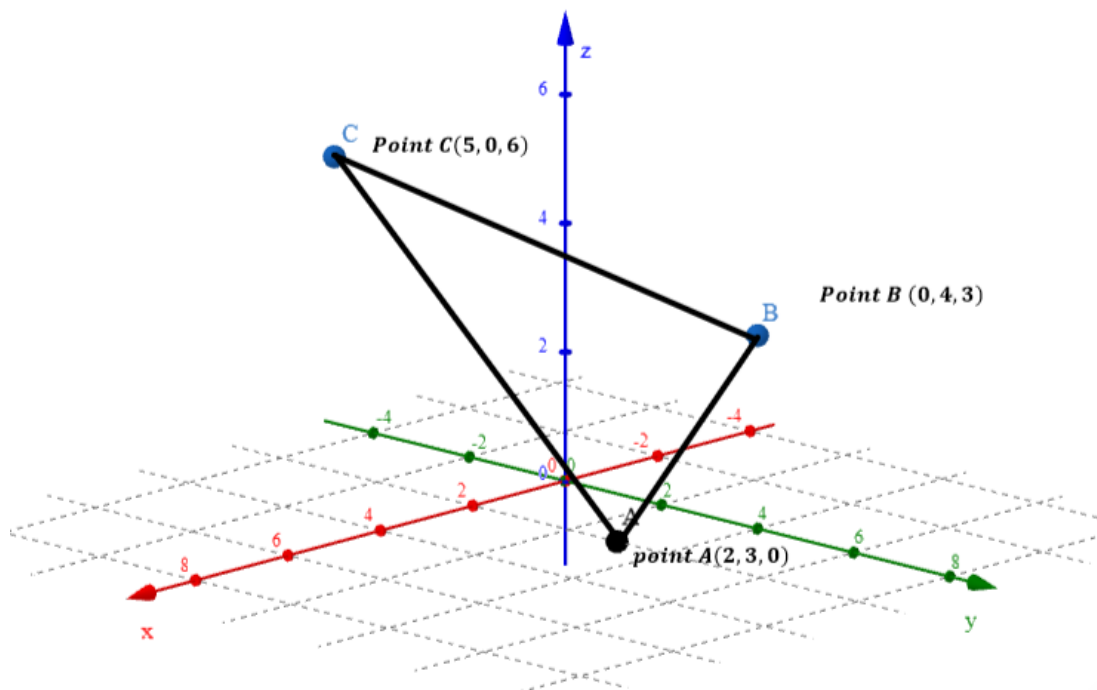
United International University
School of Science and Engineering
 Mid Term Exam Trimester: Fall 2024
 Course Title: Coordinate Geometry and Vector Analysis
 Course Code: Math 2201 Marks: 30
Total Time: 1 hour and 30 minutes

Answer all questions.

1.
 - a) Determine whether or not the given lines are intersecting at (x_0, y_0, z_0) . [3]
 $L_1: x = -1 - 4t, y = t, z = -1 - t$
 $L_2: x = 2 - 2t, y = 2 - 3t, z = -1 + 2t$
 - b) Find the equation of line of intersection of the planes [3]
 $x - y + 2z = 3$ and $x + 2y + z = 2$.
 - c) Find the vector component (orthogonal projection) of $u = \langle 0, 0, -2 \rangle$ along $w = \langle 1, 2, 0 \rangle$. [2]
 - d) Let $f(x, y) = x^2y^3 + 3$. Find the maximum value of a directional derivative at $(-1, 1)$. [2]

2.
 - a) Evaluate the iterated integral $\int_0^1 \int_0^1 \frac{xy}{\sqrt{x^2+y^2+1}} dx dy$. [5]
 - b) Using double integral to find the area enclosed by the equations [5]
 $y = -x, -x + y = 2$ and $y = 2$.

3. [10]



- a) Find the area of the triangle **ABC**
- b) Find the equation of the plane passing through the points **A, B** and **C**.
- c) What is the angle between side **AB** and **BC**?
- d) Are these three points **A(2, 3, 0), B(0, 4, 3)** and **C(5, 0, 6)** lie in same plane?

